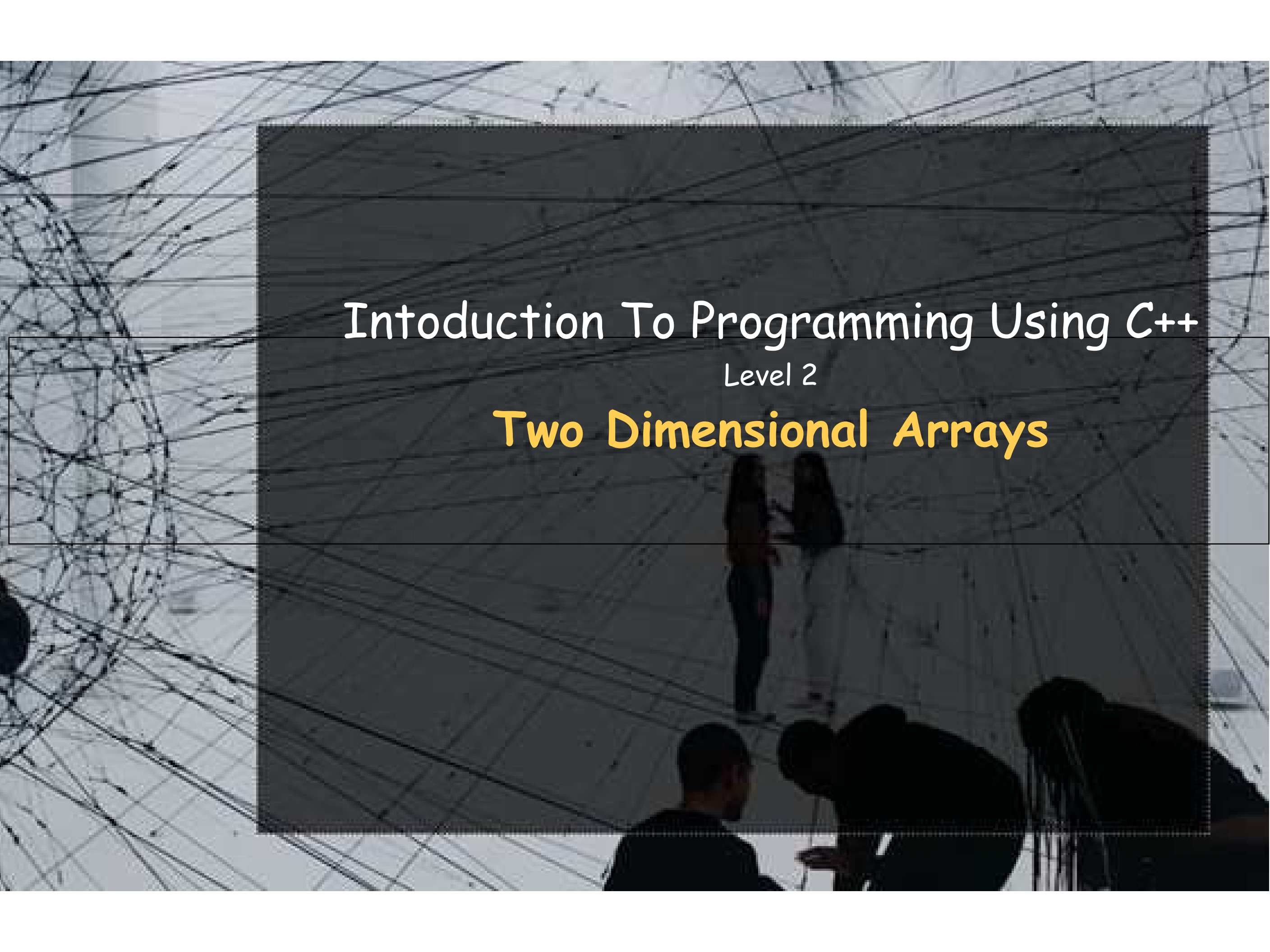




Introduction To Programming Using C++ Level 2 Two Dimensional Arrays



Introduction To Programming Using C++

Level 2

Two Dimensional Arrays

Definition

- ◆ A Two Dimensional Array is an array that contains Rows and Columns.
- ◆ It is used to store data in a table-like structure, such as a multiplication table.
- ◆ Each element is accessed using two indexes:
 `[row][col]`

Explanation

Declaration

`type arrayName[rows][cols];`

 ``rows`` = Number of Rows

 ``cols`` = Number of Columns

Data Storage in Memory

 Rows are stored one after another in contiguous memory.

 If fewer values than required are provided during initialization, the remaining elements are automatically initialized to ``0``.

 Columns are mainly a logical organization for the programmer. In memory, the elements are stored sequentially, row by row.

Procedures

1. Define the array using the number of Rows and Columns.
2. Initialize the array with values if needed.
3. Use the Row index to select a Row.
4. Use the Column index to select an element inside that Row.
5. Use nested `for` loops to access all elements.

💡 Example

```
int main()
{
    int x[3][4] =
    {
        {1, 2, 3, 4},
        {5, 6, 7, 8},
        {9, 10, 11, 12}
    };
    for (int row = 0; row < 3; row++)
    {
        for (int col = 0; col < 4; col++)
        {
            cout << x[row][col] << " ";
        }
        cout << endl;
    }
    return 0;
}
```

➡ Result

```
1 2 3 4
5 6 7 8
9 10 11 12
```

Note

- ◆ A Two Dimensional Array can have a Row containing zero values, but in a regular C++ built-in 2D array, the number of Columns must be specified.
- ◆ The Column size is part of the array's type and is required by the compiler for a built-in multidimensional array.

⊘ Warning

- Array indexes start from `0`, not `1`.
- For `int x[3][4]`:
 - ▨ Valid Row indexes: `0` to `2`
 - ▨ Valid Column indexes: `0` to `3`
- Accessing an index outside these ranges causes undefined behavior.



Comparison



One Dimensional Array

- ✓ Uses one index: ``array[index]``
- ✓ Represents a simple sequence of elements.
- ✓ Example: ``int numbers[5];``



Two Dimensional Array

- ✓ Uses two indexes: ``array[row][col]``
- ✓ Represents data organized into Rows and Columns.
- ✓ Example: ``int matrix[3][4];``

✦✦ Summary

- ✓ A Two Dimensional Array stores data using Rows and Columns.
- ✓ The syntax is ``type arrayName[rows][cols];``
- ✓ Elements are accessed using ``[row][col]``.
- ✓ Data is stored sequentially in memory, Row by Row.
- ✓ Nested ``for`` loops are commonly used to process all elements.
- ✓ Understanding ``row`` and ``col`` indexing is essential when working with Two Dimensional Arrays.

Thank you 😊